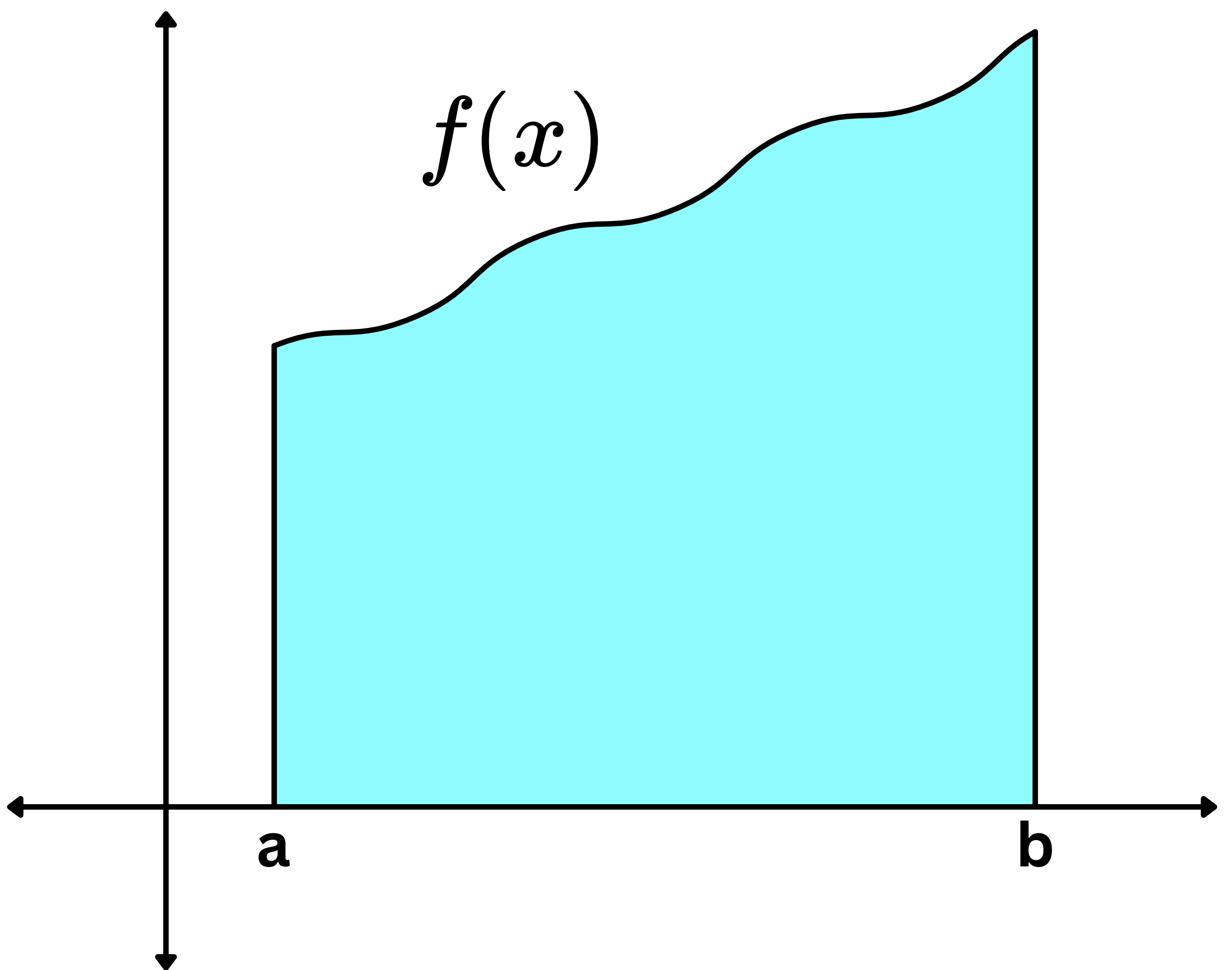
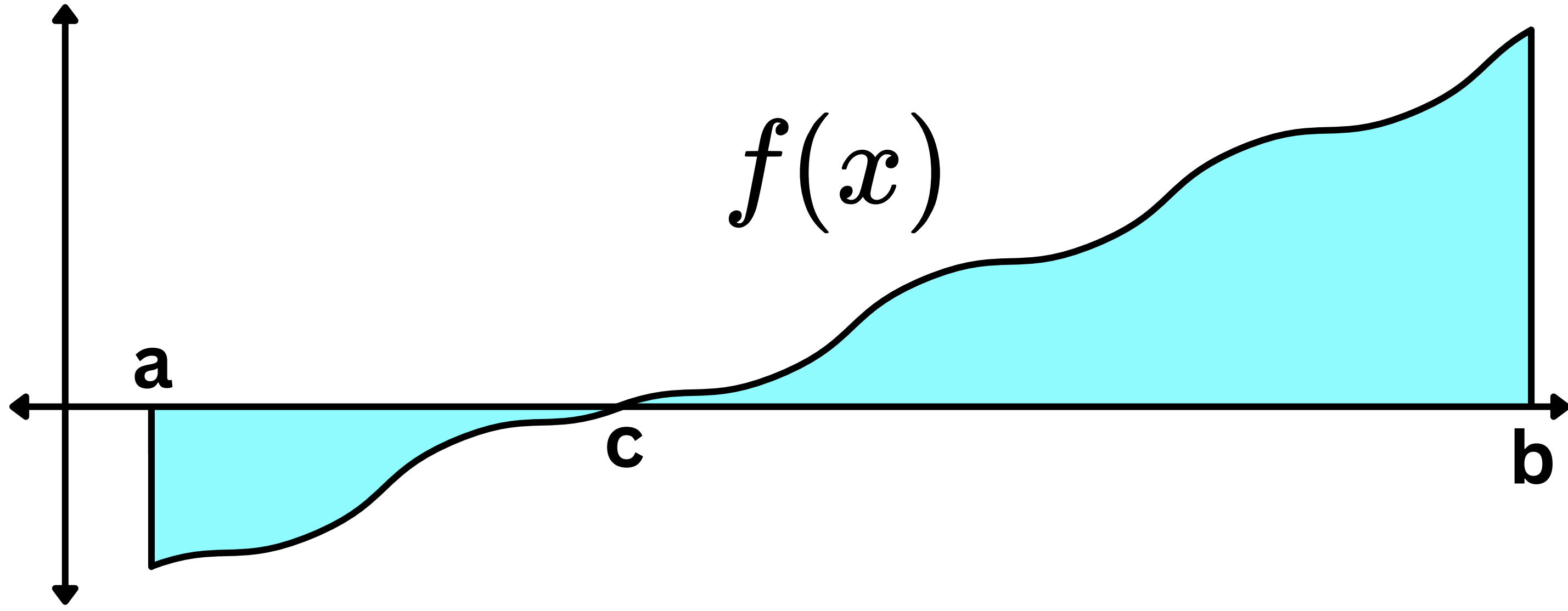


Area Under Curve

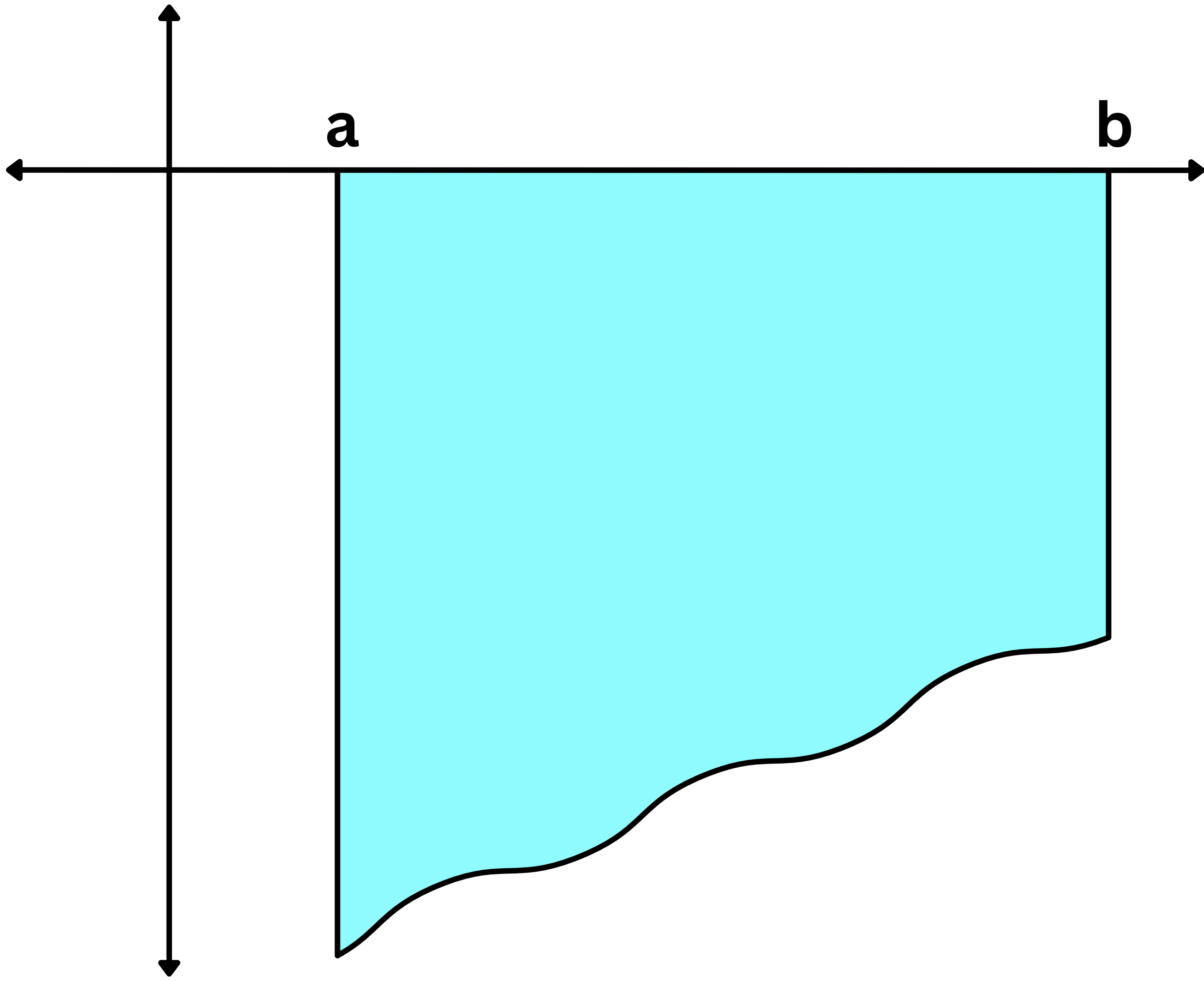
Area Bounded by a
Curve with the x-axis



$$\text{Area} = \int_a^b f(x) \, dx$$

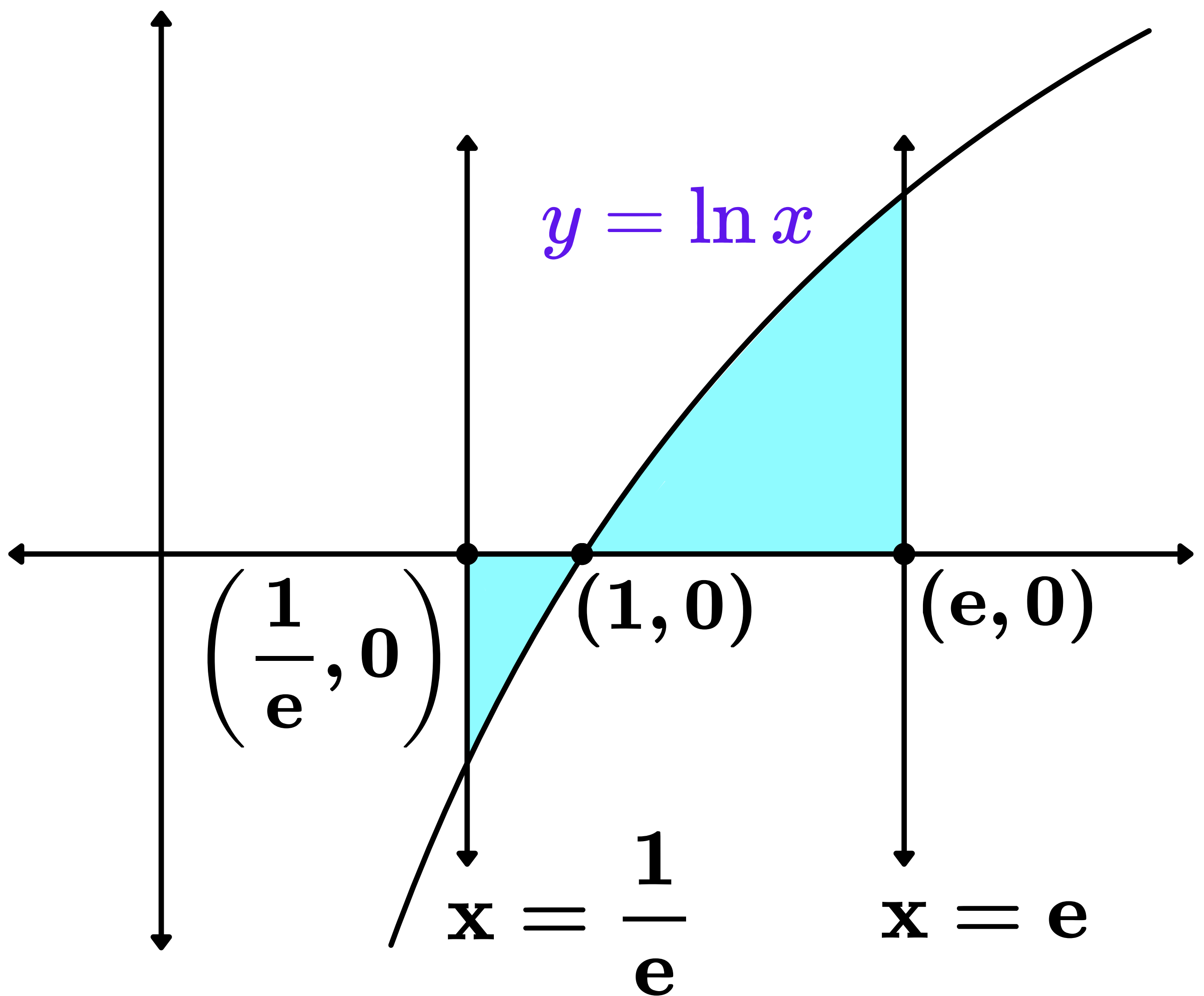


$$\text{Area} = \left| \int_a^c f(x) dx \right| + \left| \int_c^b f(x) dx \right|$$



$$\text{Area} = \left| \int_a^b f(x) \, dx \right|$$

Find the area bound by $y = \ln x$ with
 x - axis and $x = 1/e$ and $x = e$



$$\text{Area} = \left| \int_{1/e}^1 \ln x \, dx \right| + \int_1^e \ln x \, dx$$

$$= x(\ln x - 1) \Big|_{1/e}^1 + x(\ln x - 1) \Big|_1^e$$

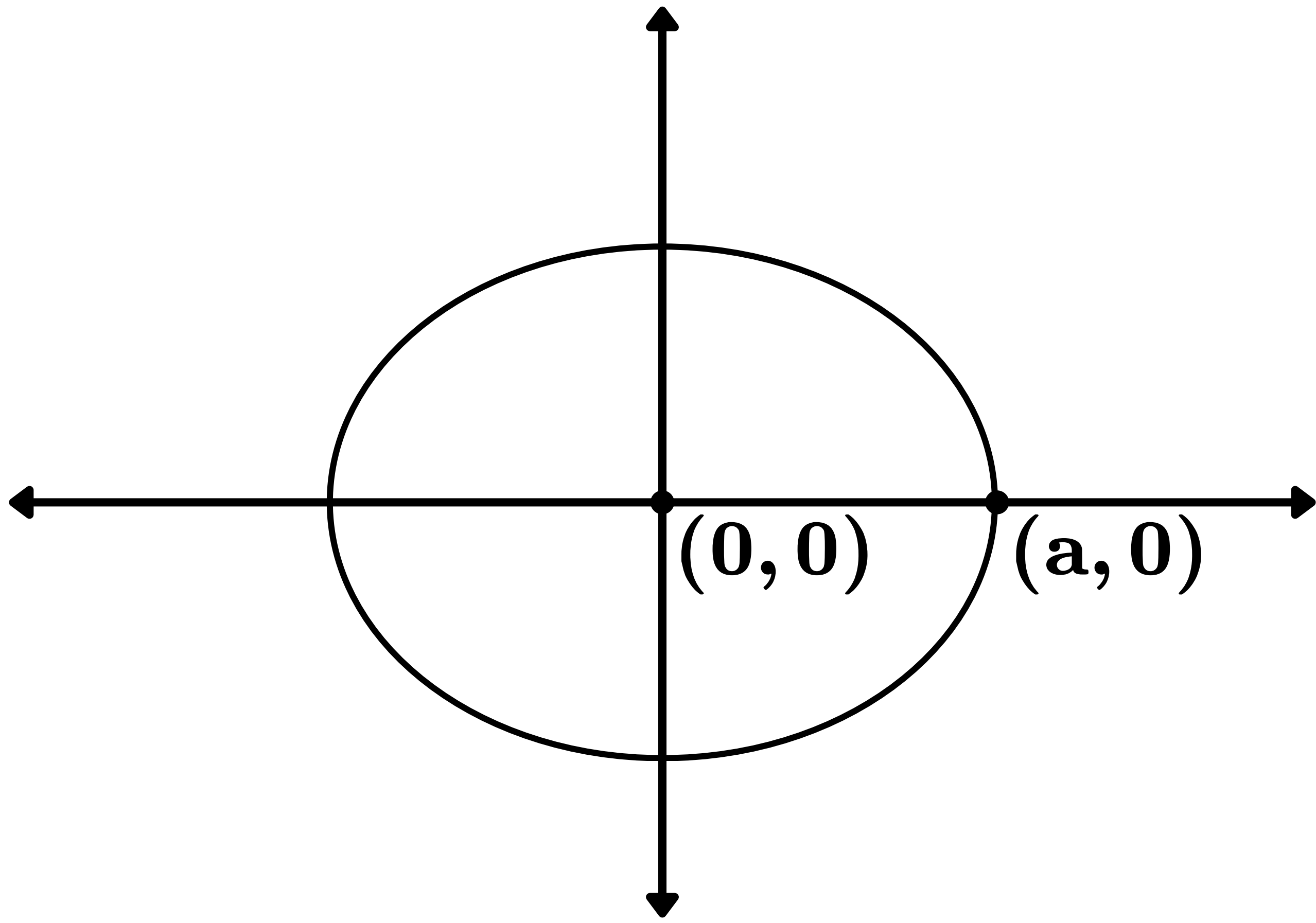
$$= 1(\ln 1 - 1) - \frac{1}{e} \left(\ln \frac{1}{e} - 1 \right)$$

$$+ e(\ln e - 1) - 1(\ln 1 - 1)$$

$$= -1 - \frac{1}{e}(-1 - 1) + 1$$

$$= -\left(-1 + \frac{2}{e}\right)$$

$$\left(1 - \frac{2}{e}\right) + 1 = \left(2 - \frac{2}{e}\right)$$



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{y^2}{b^2} = 1 - \frac{x^2}{a^2}$$

$$y = \frac{b}{a} \sqrt{a^2 - x^2}$$

The Area Enclosed by the Ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ is } \pi ab \rightarrow$$

$$\Rightarrow \text{Area} = 4 \int_0^a y \, dx$$

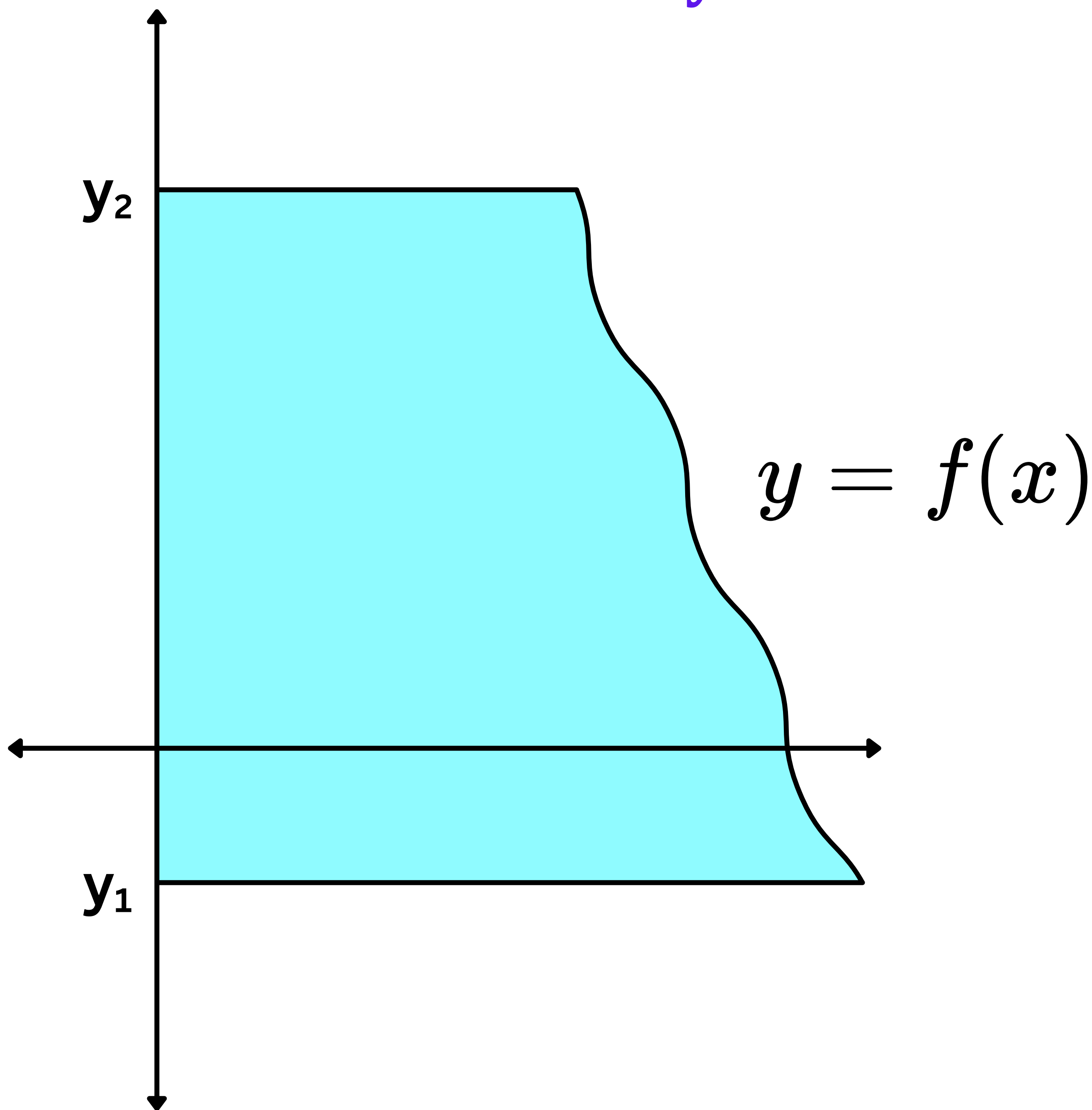
$$= 4 \int_0^a \frac{b}{a} \sqrt{a^2 - x^2} \, dx$$

$$= \frac{4b}{a} \int_0^a \sqrt{a^2 - x^2} \, dx$$

$$= \frac{4b}{a} \left\{ \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} \right\}_0^a$$

$$= \frac{4b}{a} \left\{ \frac{a^2}{2} \sin^{-1} 1 \right\} = \frac{4ab}{2} \cdot \frac{\pi}{2} = \pi ab$$

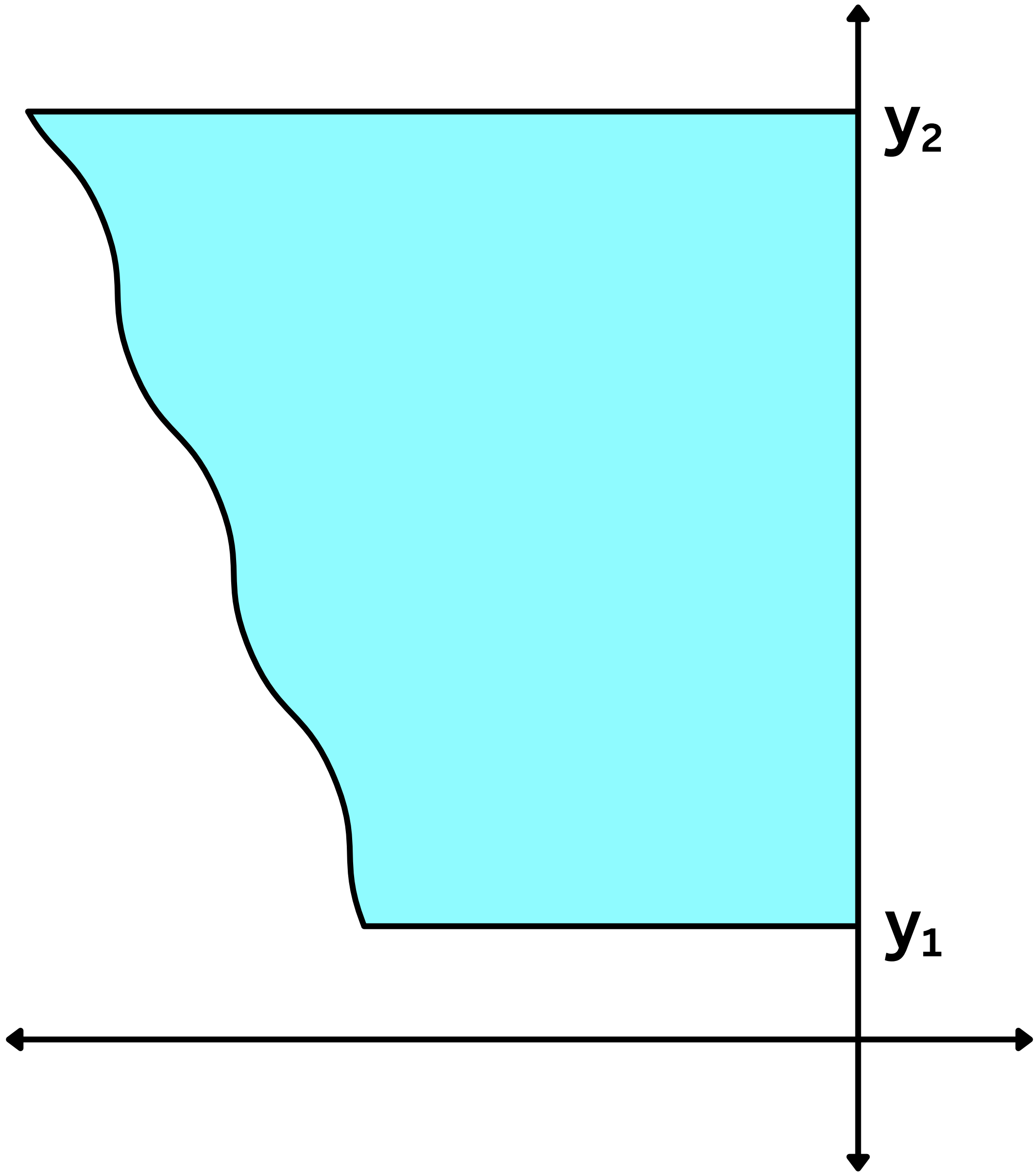
Area Bounded by a Curve with the y-axis



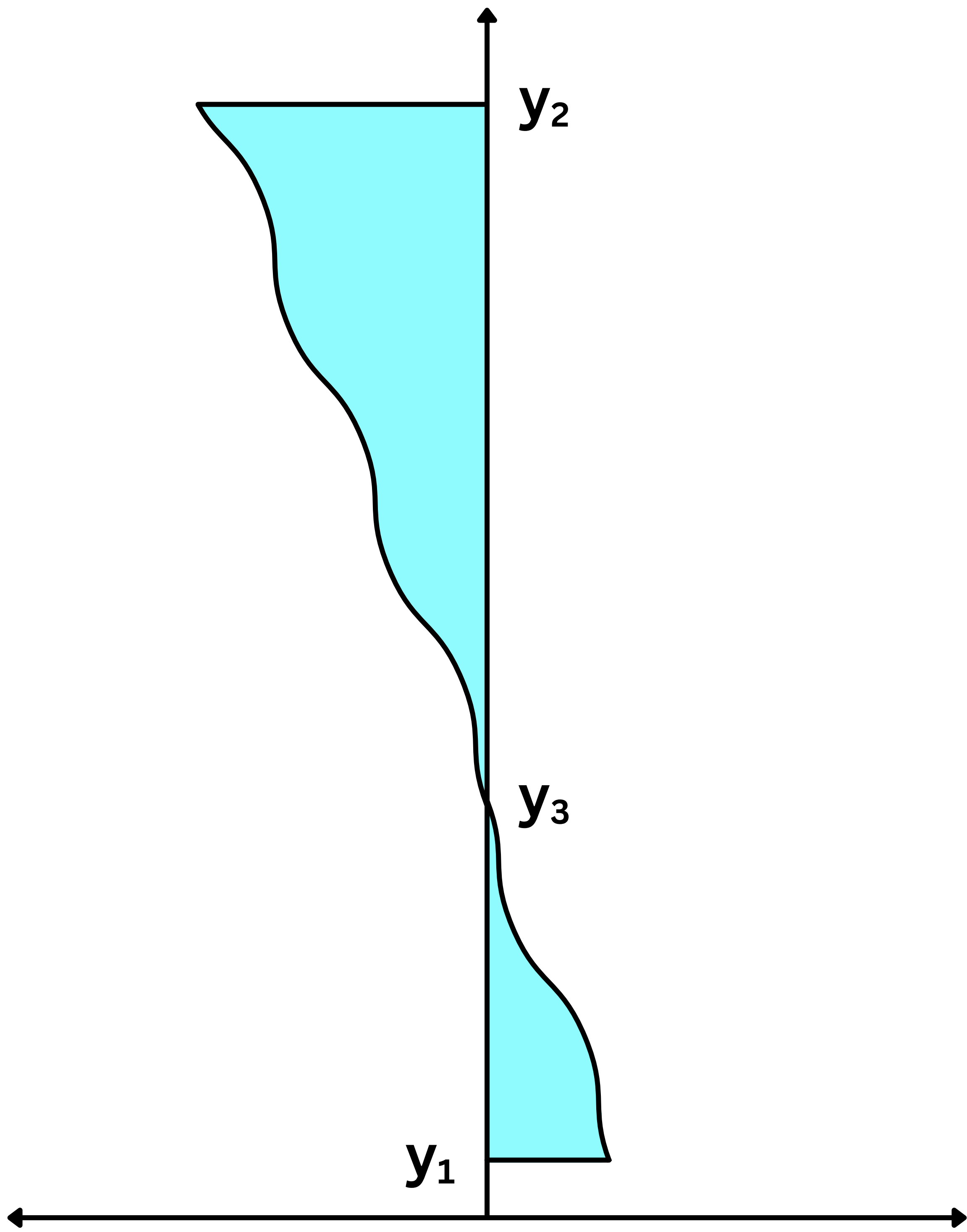
$$y = f(x)$$

$$x = f^{-1}(y)$$

$$\text{Area} = \int_{y_1}^{y_2} x \, dy = \int_{y_1}^{y_2} f^{-1}(y) \, dy$$

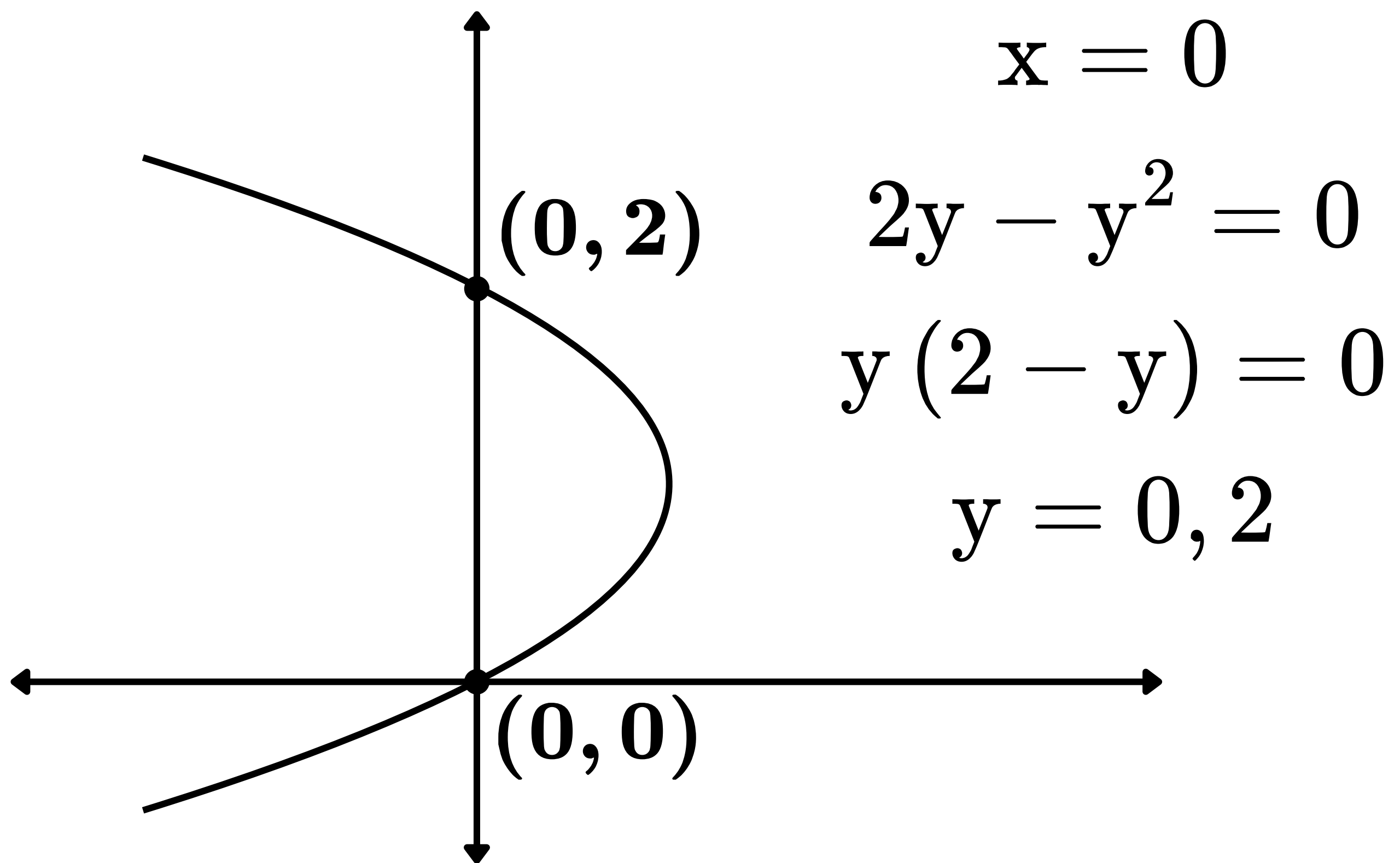


$$\text{Area} = \left| \int_{y_1}^{y_2} x \, dy \right|$$



$$\text{Area} = \int_{y_1}^{y_3} x \, dy + \left| \int_{y_3}^{y_2} x \, dy \right|$$

**Area bounded by $x = 2y - y^2$
and y -axis $\rightarrow$**



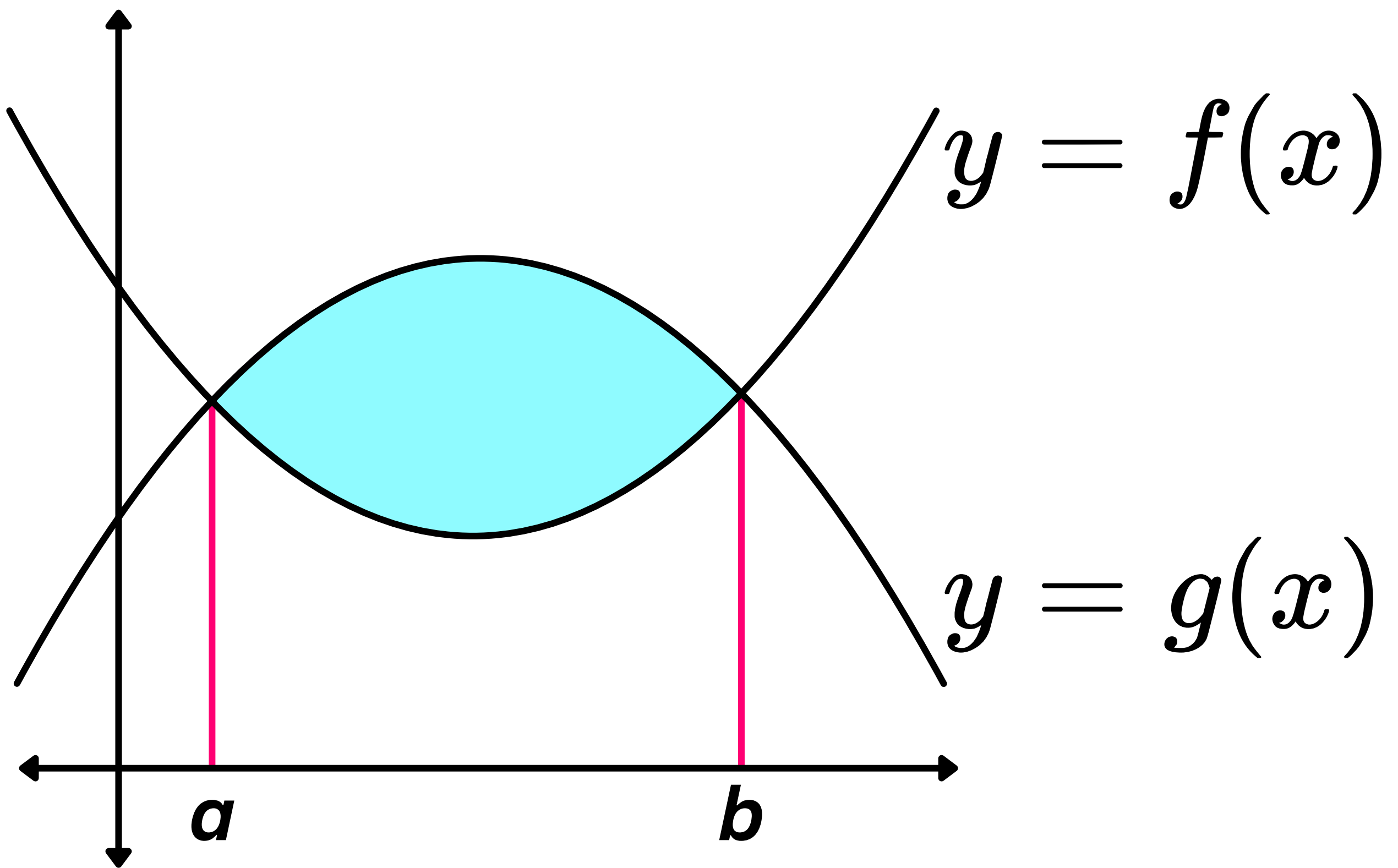
$$\text{Area} = \int_0^2 x \, dy$$

$$= \int_0^2 (2y - y^2) \, dy$$

$$= \left[y^2 - \frac{y^3}{3} \right]_0^2 = \left(2^2 - \frac{8}{3} \right) - 0$$

$$4 - \frac{8}{3} = \frac{12 - 8}{3} = \frac{4}{3}$$

Area Bounded Between 2 Curves w.r.t to the x-axis



Step 1: Find the points of intersection of the curves.

$$\begin{aligned}\text{Area} &= \left| \int_a^b f(x) \, dx - \int_a^b g(x) \, dx \right| \\ &= \left| \int_a^b (f(x) - g(x)) \, dx \right|\end{aligned}$$

Q: The area of the region enclosed by the curve $y = x^3$ and its tangent at the point $(-1, -1)$ is ____.

$$\frac{dy}{dx} = 3x^2 = 3(-1)^2 = 3$$

$$m_T = 3$$

Point $(-1, -1)$

Equation of Tangent:

$$y + 1 = 3(x + 1)$$

$$y = 3x + 2$$

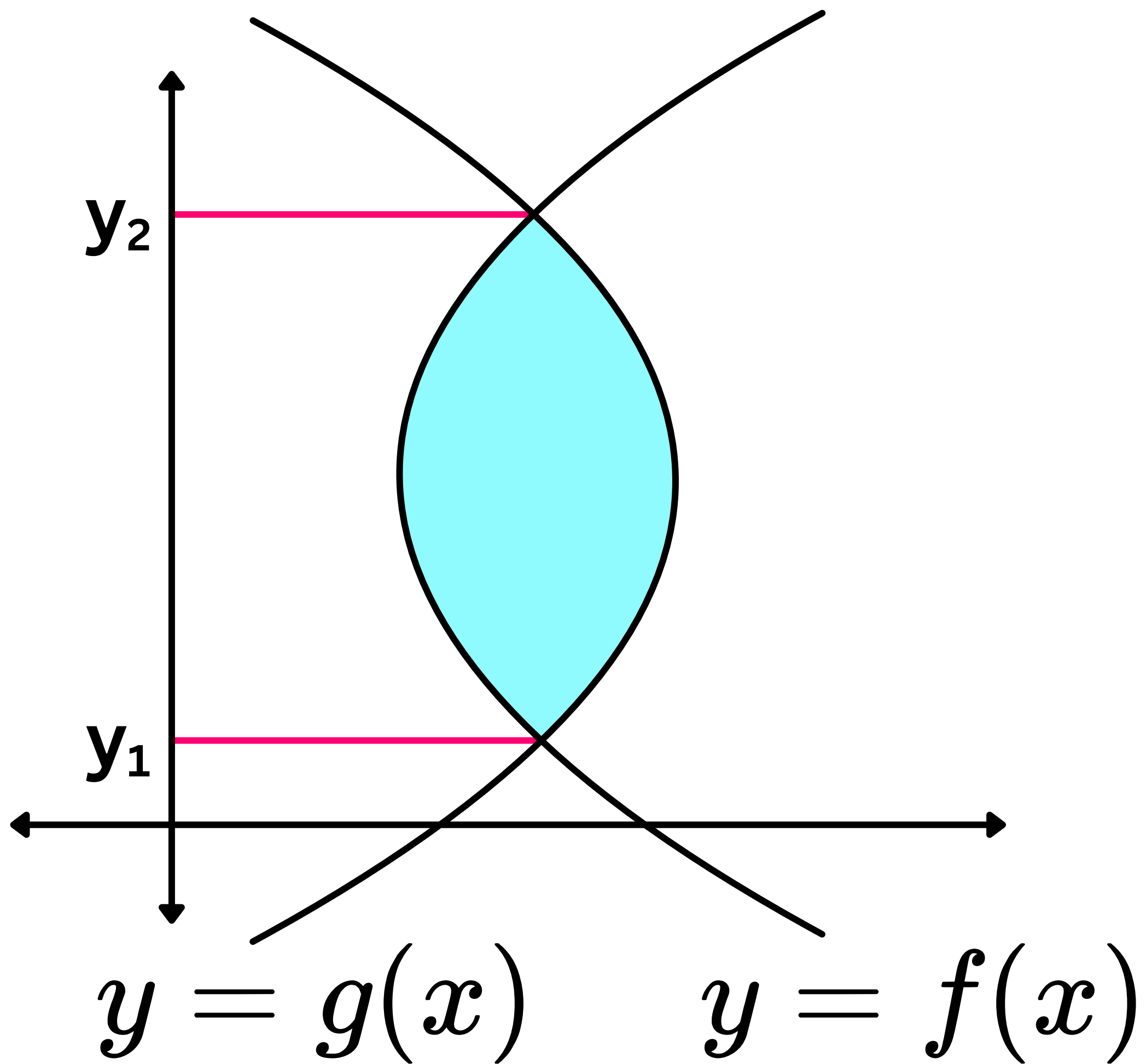
$$x^3 = 3x + 2$$

$$x = 2, -1$$

$$\text{Area} = \left| \int_{-1}^2 ((3x + 2) - x^3) dx \right|$$

$$= \frac{3x^2}{2} + 2x - \frac{x^4}{4} \Big|_{-1}^2 = \frac{27}{4}$$

Area Bounded Between 2 Curves w.r.t to the y-axis



$$\left| \int_{y_1}^{y_2} x_f dy - \int_{y_1}^{y_2} x_g dy \right|$$

$$\text{Area} = \left| \int_{y_1}^{y_2} (x_f - x_g) dy \right|$$

Important Formulas for Area Bound Between 2 Curves

The area bounded by $y^2 = 4ax$ & $x^2 = 4by$ is $\frac{16ab}{3}$

The area bounded by $y^2 = 4ax$ & $y = mx$ is $\frac{8a^2}{3m^3}$

The area bounded by $x^2 = 4ay$ & $y = mx$ is $\frac{8a^2m^3}{3}$